

ABSTRACT

The Medicine Reminder System is an IoT-based embedded project designed to help patients, elderly individuals, and caregivers manage medication schedules effectively. The system uses an ESP32 microcontroller as the central processing unit, integrated with a DS3231 RTC module for accurate timekeeping, a 16x2 LCD display for real-time status output, LEDs for visual alerts, an active buzzer for audio reminders, and push buttons for user interaction. When the scheduled medicine time arrives, the buzzer sounds and the corresponding LED lights up to indicate which medicine box needs to be taken. The user confirms by pressing the OK button. Additionally, the ESP32's built-in Wi-Fi capability enables a web-based interface, allowing remote monitoring of medication schedules from any connected device. The system is programmed using the Arduino IDE with libraries including Wire.h, LiquidCrystal_I2C.h, RTCLib.h, WiFi.h, and WebServer.h. This project demonstrates the practical application of IoT in healthcare, providing a low-cost, reliable, and user-friendly solution for medication adherence.

CONTENTS

SI No	Content	Page No
1	Introduction	4
2	Components and Specifications	5-8
3	System Design and Architecture	9-10
4	Implementation	11-12
5	Applications	13
6	Future Enhancements	14
7	Conclusion	15

1. INTRODUCTION

1.1 Background

Medication non-adherence is a significant global health challenge. According to the World Health Organization, approximately 50% of patients with chronic diseases do not take their medications as prescribed. This leads to disease progression, increased hospitalizations, and preventable deaths. In elderly populations and households with multiple patients managing different medicines at different times of the day, the risk of missed or incorrect dosing is even higher.

With the rapid advancement of the Internet of Things (IoT), embedded systems can now be designed to automate daily healthcare tasks, including medication reminders. IoT-based health monitoring devices are increasingly being adopted in smart homes, hospitals, and personal care environments. These devices provide real-time alerts, remote monitoring, and reliable scheduling that traditional alarm-based methods cannot offer.

1.2 Objective

The primary objective of this project is to design and implement a Medicine Reminder System using the ESP32 microcontroller that:

- Tracks real-time using a DS3231 RTC module with battery backup.
- Displays the current time and medicine alert status on a 16x2 LCD screen.
- Triggers visual (LEDs) and audio (buzzer) alerts at pre-set medicine times.
- Allows users to confirm medicine intake using push buttons.
- Provides a Wi-Fi-based web interface for remote monitoring of medication schedules.

1.3 Scope

This project is scoped for individual or small household use. The system supports three separate medicine slots (Box 1, Box 2, Box 3), each assigned a unique LED color for easy identification. The reminder time can be configured using the onboard push buttons. The system is powered via a standard 5V USB supply, making it portable and easy to use. The web interface, hosted on the ESP32, can be accessed over a local Wi-Fi network from any browser-enabled device such as a smartphone or laptop.

2. COMPONENTS AND SPECIFICATIONS

This section describes each hardware and software component used in the Medicine Reminder System along with its role and technical specifications.

2.1 Microcontroller – ESP32 Dev Board

The ESP32 is the main controller of the system. It is a dual-core, 32-bit microcontroller with integrated Wi-Fi and Bluetooth capabilities. Its processing power and connectivity features make it ideal for IoT applications.

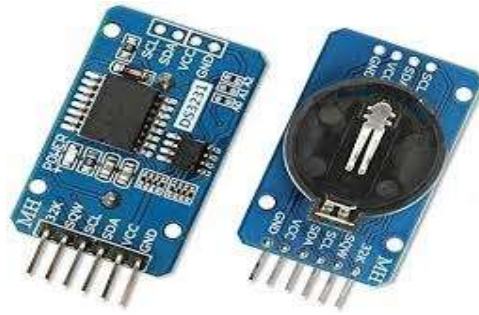
Parameter	Specification
Processor	Xtensa dual-core 32-bit LX6 @ up to 240 MHz
Flash Memory	4 MB
RAM	520 KB SRAM
Wi-Fi	802.11 b/g/n (2.4 GHz)
GPIO Pins	30+ (supports I2C, SPI, UART, PWM)
Operating Voltage	3.3V logic, 5V USB power
Programming	Arduino IDE / MicroPython



2.2 Time Module – DS3231 RTC Module

The DS3231 is a high-precision Real-Time Clock (RTC) module that maintains accurate time even when the main power is disconnected, thanks to a backup coin-cell battery. It communicates with the ESP32 via the I2C protocol.

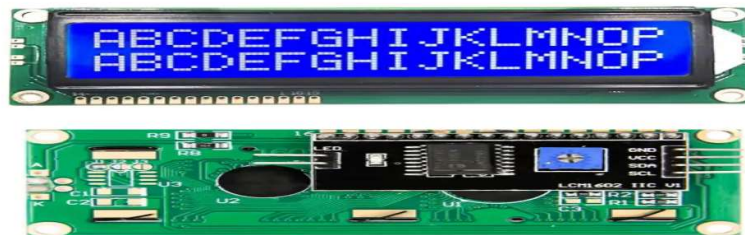
- Interface: I2C (SDA and SCL pins)
- Accuracy: ± 2 ppm from 0°C to +40°C
- Battery backup: CR2032 coin cell (retains time during power loss)
- Provides: Hours, Minutes, Seconds, Day, Date, Month, Year



2.3 Display – 16x2 LCD with I2C Module

A standard 16x2 character LCD display is used to show the current time and medicine reminder status. The I2C backpack module (PCF8574) reduces the number of GPIO pins required from 6–10 to just 2 (SDA and SCL), keeping the wiring simple.

- Display Size: 16 columns x 2 rows
- Interface: I2C via PCF8574 I/O expander
- Operating Voltage: 5V
- Backlight: LED backlight for low-light visibility



2.4 Output Devices

The system uses the following output devices to alert the user:

- Active Buzzer (e.g., SEM 1201): Generates a continuous alarm sound when a medicine reminder is triggered. Operating voltage: 5V. Sound output: ~85 dB.
- Red LED: Indicates Medicine Box 1 reminder.
- Blue LED: Indicates Medicine Box 2 reminder.
- Green LED: Indicates Medicine Box 3 reminder.

Each LED is connected to a dedicated GPIO pin on the ESP32 through a 220 Ω current-limiting resistor.



2.5 Input Devices – Push Buttons

Three push buttons are used to interact with the system:

- SET Button: Enters the time-setting mode.
- INC Button: Increments the time value (hours/minutes) during setting.
- OK Button: Confirms the selected value or acknowledges that a medicine has been taken.

Pull-up resistors ($10k\Omega$) may be used externally or the ESP32's internal pull-up resistors can be enabled in software.



2.6 Passive Components

Component	Value / Quantity	Purpose
Resistors	$220\Omega \times 3$	Current limiting for LEDs
Resistors	$10k\Omega \times 3$ (optional)	Pull-down/pull-up for push buttons
Breadboard	1	Prototyping and wiring
Jumper Wires	As required	Connections between components
USB Cable	5V	Power supply for ESP32



2.7 Software and Libraries

The system is programmed using the Arduino IDE. The following libraries are used:

- Wire.h – Enables I2C communication between ESP32, DS3231 RTC, and LCD.
- LiquidCrystal_I2C.h – Provides functions to control the 16x2 LCD via I2C.
- RTCLib.h – Manages RTC operations such as reading and setting time/date.
- WiFi.h – Enables ESP32 to connect to a Wi-Fi network.
- WebServer.h – Hosts a simple HTTP web server on the ESP32 for the web interface.

3. SYSTEM DESIGN AND ARCHITECTURE

3.1 System Overview

The Medicine Reminder System is designed around the ESP32 microcontroller, which acts as the central hub connecting all peripherals. The DS3231 RTC continuously tracks time and provides accurate timestamps to the ESP32. The ESP32 compares the current time with the stored medicine schedule and triggers the appropriate reminder.

The system operates in three modes:

- Normal Mode: Displays current time on LCD and monitors the schedule.
- Setting Mode: Allows the user to configure reminder times using push buttons.
- Reminder Mode: Activates buzzer and corresponding LED when medicine time arrives.

3.2 Circuit Diagram

The circuit consists of the following connections:

- DS3231 RTC: SDA → GPIO 21, SCL → GPIO 22 (I2C bus)
- 16x2 LCD (I2C): SDA → GPIO 21, SCL → GPIO 22 (shared I2C bus, different address)
- Active Buzzer: Signal pin → GPIO 26
- Red LED (Box 1): Anode → GPIO 25 via 220Ω resistor → GND
- Blue LED (Box 2): Anode → GPIO 27 via 220Ω resistor → GND
- Green LED (Box 3): Anode → GPIO 14 via 220Ω resistor → GND
- SET Button: One terminal → GPIO 32, other terminal → GND (internal pull-up enabled)
- INC Button: One terminal → GPIO 33, other terminal → GND (internal pull-up enabled)
- OK Button: One terminal → GPIO 35, other terminal → GND (internal pull-up enabled)

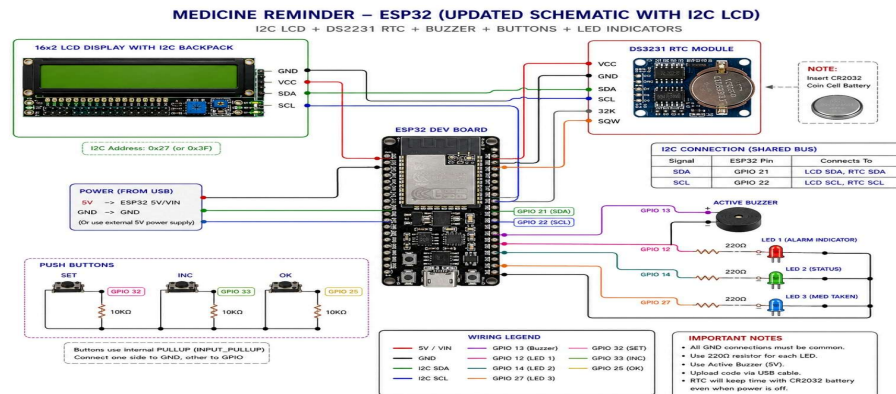


Figure 1: Circuit Diagram of the Medicine Reminder System

3.3 Data Flow

The data flow of the system is as follows:

- The DS3231 RTC module continuously maintains the current date and time.
- The ESP32 reads the current time from the RTC via the I2C bus every second.
- The current time is displayed on the 16x2 LCD.
- The ESP32 compares the current time with the three stored medicine alarm times.
- On a time match, the buzzer is activated and the corresponding LED lights up.
- The user presses the OK button to acknowledge the reminder, turning off the buzzer and LED.
- The reminder status (taken/not taken) is updated and accessible via the web interface.
- The ESP32 hosts a web server accessible via its IP address on the local Wi-Fi network.

4. IMPLEMENTATION

4.1 Assembly

The hardware assembly was performed on a breadboard for prototyping. The ESP32 Dev Board was placed at the center of the breadboard. The DS3231 RTC and the 16x2 LCD (both using I2C) were connected to the shared I2C bus (GPIO 21 for SDA and GPIO 22 for SCL). The active buzzer and three LEDs were connected to their respective GPIO pins with appropriate resistors. The three push buttons were connected with the ESP32's internal pull-up resistors enabled in firmware.

4.2 Software

The firmware was developed in the Arduino IDE with the ESP32 board package installed. The program flow is as follows:

- Setup Phase: Initialize I2C, LCD, RTC, pin modes for LEDs, buzzer, and buttons. Connect to Wi-Fi and start the web server.
- Main Loop: Read current time from DS3231. Display on LCD. Check against alarm times. If match: turn on buzzer and LED. Poll for OK button press to dismiss alarm.
- Web Interface: The WebServer library serves an HTML page showing the current time, medicine schedule, and acknowledgment status for each box.

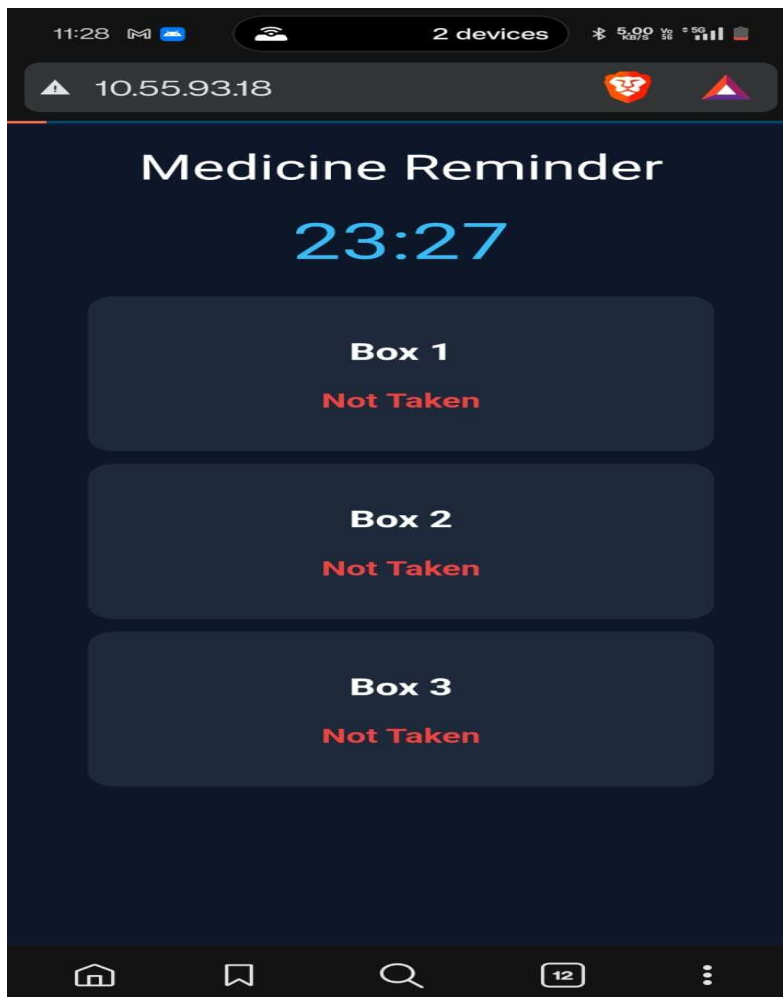
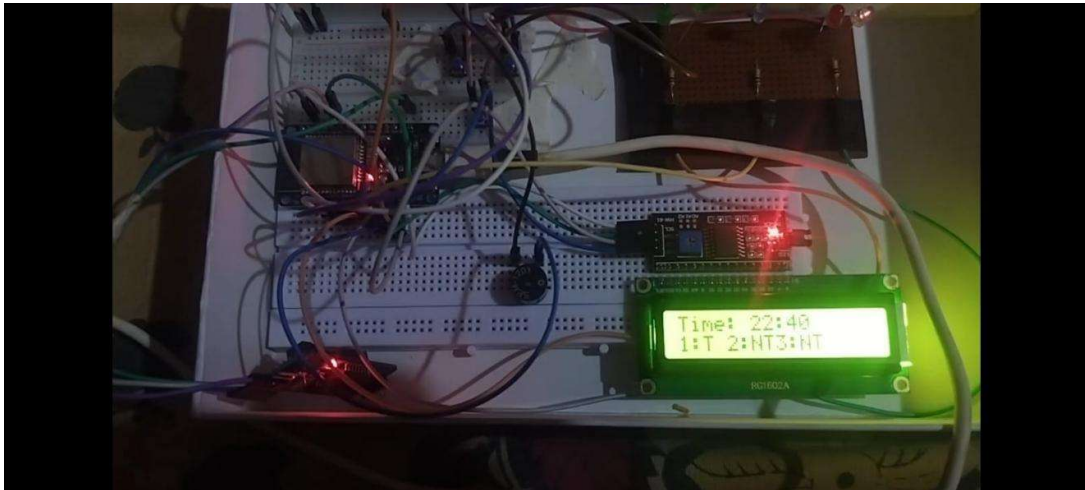
The SET and INC buttons allow the user to navigate through hour and minute settings for each of the three alarm slots. The OK button confirms each value during the setting process.

4.3 Testing

The system was tested under the following conditions:

- Correct time display on the LCD after RTC initialization.
- Alarm trigger accuracy: buzzer and LED activated within 1 second of scheduled time.
- Button response: OK button correctly dismisses the alarm.
- Battery backup: time retained correctly after removing main power.
- Web interface: accessible from a smartphone browser on the same Wi-Fi network.
- Multiple alarms: all three medicine slots tested independently without conflict.

4.4 Results



5. APPLICATIONS

The Medicine Reminder System has a wide range of real-world applications:

- **Elderly Care:** Assists elderly individuals who may forget their daily medications, especially those with multiple prescriptions at different times.
- **Chronic Disease Management:** Useful for patients with diabetes, hypertension, cardiac conditions, or other chronic diseases requiring strict medication schedules.
- **Hospital Outpatient Monitoring:** Can be deployed in outpatient settings to remind patients about post-discharge medicines.
- **Caregiver Support:** Enables caregivers to remotely monitor whether a patient has taken their medicine via the Wi-Fi web interface.
- **Smart Home Integration:** Can be integrated with smart home ecosystems (e.g., Google Home, Alexa) for voice-based reminders.
- **Pediatric Use:** Parents can use the system to ensure children take their prescribed medicines on time.
- **Pharmacy Compliance Programs:** Pharmacies can embed similar systems in smart pill dispensers to improve patient adherence.

6. FUTURE ENHANCEMENTS

The following enhancements can be incorporated in future versions of the system:

- **Mobile App Integration:** Develop a dedicated Android/iOS app for push notifications and schedule management, replacing the basic web interface.
- **Cloud Connectivity:** Integrate with cloud platforms such as AWS IoT, Google Firebase, or ThingSpeak for remote data logging and analytics.
- **Pill Dispenser Mechanism:** Add a servo motor-based dispensing mechanism to automatically release the correct pill at the scheduled time.
- **Voice Alerts:** Incorporate a speaker module (e.g., DFPlayer Mini) to announce medicine reminders in the user's language.
- **Multi-user Support:** Expand the system to handle multiple users, each with independent schedules, managed via the web interface.
- **Battery-powered Operation:** Add a Li-Po battery and charging circuit for portability and use during power outages.
- **SMS Alerts:** Use a GSM module (e.g., SIM800L) to send SMS alerts if medicine is not acknowledged within a set time limit.
- **OLED Display Upgrade:** Replace the 16x2 LCD with an OLED display for improved readability and aesthetic appeal.

7. CONCLUSION

The Medicine Reminder System successfully demonstrates the application of IoT technology in the healthcare domain. By integrating the ESP32 microcontroller with a DS3231 RTC module, a 16x2 LCD, active buzzer, LEDs, and push buttons, the system provides a reliable and cost-effective solution for medication adherence.

The system ensures that users are reminded about their medicines at the correct times through both visual and audio alerts. The acknowledgment mechanism prevents reminders from being ignored, and the Wi-Fi-based web interface enables caregivers to monitor medicine intake remotely. The use of widely available, affordable components and the Arduino IDE makes this project accessible for educational purposes and practical deployment.

This project not only addresses a critical public health issue but also showcases how embedded IoT systems can be designed to improve daily quality of life. With the future enhancements proposed, this system has the potential to evolve into a comprehensive smart medication management device suitable for individual, institutional, and commercial use.